

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who delivers data-centric integrations and production AI for regulated enterprise customers. Data-pipeline depth from Emerson Synthesis Order-to-Cash - T-SQL stored procedures, Jenkins CI/CD with rollback checkpoints, Grafana dashboards, reconciliation queries, RBAC + audit logging in a compliance-sensitive financial-modules environment. Production AI at Progress: Healthcare RAG (~35% retrieval precision), agentic LLM workflows (>30% hallucination reduction), predictive ML pipelines (15-20% recall lift on high-risk categories), FastAPI / Docker production (~30% defect reduction). Manga Lens solo-shipped to Chrome Web Store with four AI vision providers - direct integration breadth.

SKILLS

Forward Deployed Engineering • Data Pipeline Design • Enterprise Integration Workflows • Python / TypeScript / SQL •
Production RAG & Agents • Cloud-Ready Deployment • CI/CD Discipline • Compliance & Audit Logs

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

Data Pipelines & Analytics: T-SQL stored procedures, SQL Server DMVs, ETL preprocessing (Pandas, NumPy), reconciliation queries, partition-aware indexing, walk-forward validation, feature engineering, time-series forecasting

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins CI/CD with rollback checkpoints, Grafana, RESTful APIs, structured logging, observability, MLOps/LLMOps, cloud-ready deployment, RBAC, audit logging

Domains: Healthcare AI (HIPAA-conscious), Financial-Modules / Order-to-Cash (Oil & Gas), Applied AI, Enterprise AI

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Built **data pipelines and preprocessing** (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; dataset reliability >98%, downstream model instability ~40%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, support prioritization; recall +15-20% on high-risk categories at precision >90%.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings; contextual retrieval precision +~35%, grounded-response alignment >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, structured logging + load simulation; post-deployment defects ~30%.
- Maintained HIPAA-conscious data governance: de-identification, lineage documentation, evaluation audit trails, stakeholder-facing system-limitation docs - directly transferable to financial-services compliance contexts.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India (Oil & Gas, Enterprise ERP, Financial Modules)

- Optimized SQL and T-SQL stored procedures powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); query execution $\sim 20\%$, data retrieval latency $\sim 25\%$.
- Built CI/CD pipelines with Jenkins for schema updates, version-controlled SQL scripts, stored procedure deployments with rollback checkpoints; deployment errors $\rightarrow 30\%$, release cycle $\sim 35-40\%$.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; incident recurrence $\sim 25\%$, root-cause resolution $\sim 30\%$.
- Tuned database maintenance and reconciliation queries for financial modules; reconciliation runtime $\sim 15\%$, deadlocks $\sim 15\%$.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive oil & gas environment - directly transferable to asset-management compliance posture.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India

- Built a transformer-based video summarization system; review time $\sim 60-70\%$; $\sim 85\%$ highlight alignment.
- Implemented document Q&A with RAG; hallucinated outputs $\sim 30\%$, answer accuracy $> 85\%$.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) with per-provider payload handling. Direct multi-provider integration evidence.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline with schema-validated JSON contracts between agents. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces - same architectural pattern transfers to financial-services fraud and reconciliation audits.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs. Intermediate structured prompt transformation layers improved contextual alignment $\sim 30\%$ and first-pass image success rate $\sim 25-30\%$.

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a unified YOLOv8 model; inference latency $\sim 30\%$; false positives $\sim 25\%$.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025